



The RMS



DMS



DDS



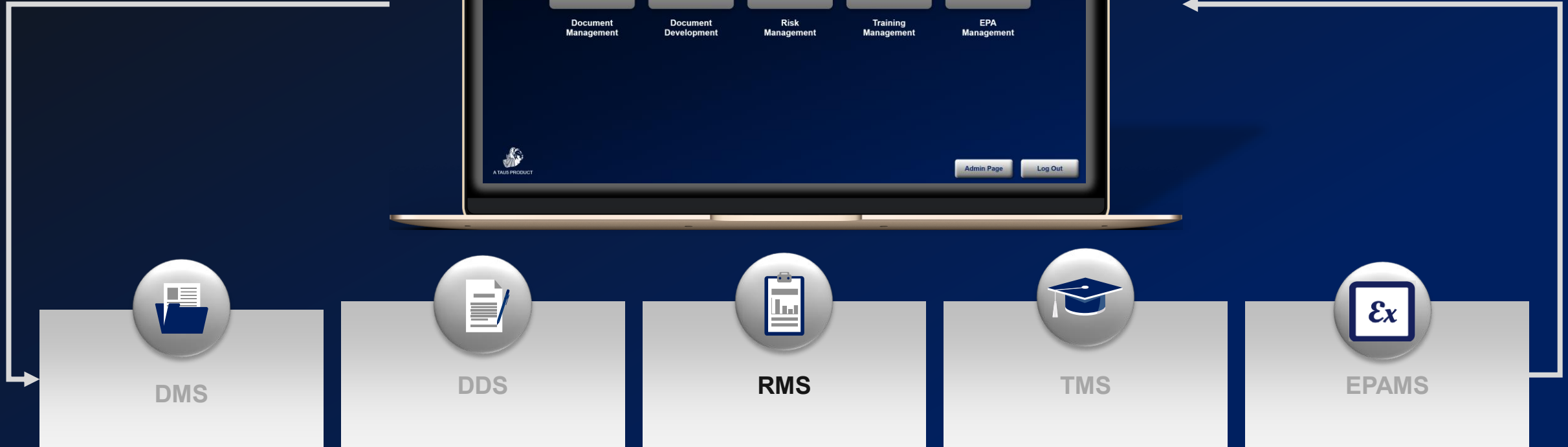
RMS



TMS

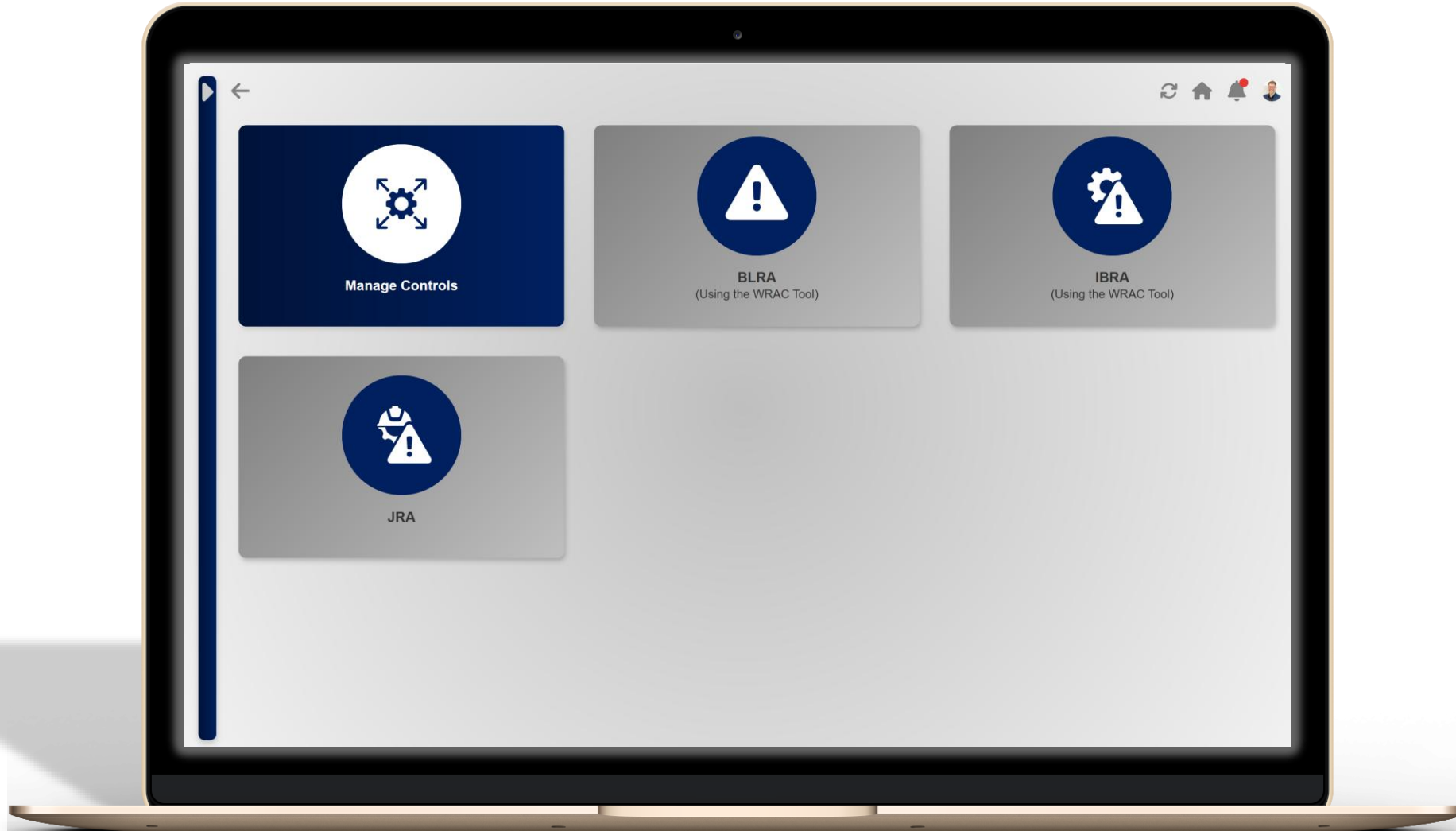


EPAMS





# Risk Management System (RMS)





# Risk Management System Description

The laptop screen shows the 'Issue Based Risk Assessment (IBRA)' tool. It features a table with four rows of risk assessments. Each row includes an 'Unwanted Event', 'Current Controls', 'Likelihood of UE', and a 'Max Risk Rank' (PUE and MUE). The table is color-coded: yellow for high risk, green for medium risk, and blue for low risk.

| Nr. | Unwanted Event                                                                                                   | Current Controls                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Likelihood of UE | (S)     | (L&R)  | (M)    | Max Risk Rank | PUE | MUE |
|-----|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|--------|--------|---------------|-----|-----|
| 1   | TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation | <ul style="list-style-type: none"><li>Competency-based training (for TMM operators)</li><li>CPS L9 SF - Detection and intervention system fail to safe (FTS) design</li><li>CPS L9 SF - Ramp Assist (RA) functionality</li><li>Traffic management system - brake test ramps and testing points</li><li>Traffic management system - mine access and mine access routes</li><li>Hardware &amp; firmware installation &amp; configuration documentation (for CPS)</li><li>Formal inspection and preventative/ condition-based maintenance system (for TMM)</li><li>Formalised pre-use inspections</li><li>Add Periodic UAT for detection accuracy and intervention capability. Define control details</li></ul> | 2. Unlikely      | 4. High | -      | -      | 14 (S)        | Y   | N   |
| 2   | Personnel bypasses CPS safety functionality                                                                      | <ul style="list-style-type: none"><li>CPS L9 SF - Supplier FMECA and product risk assessment</li><li>CPS L9 SF - Detection and intervention system fail to safe (FTS) design</li><li>Hardware &amp; firmware installation &amp; configuration documentation (for CPS)</li><li>Software version control and release management (for CPS)</li><li>Supplier warranty and performance obligation (for CPS)</li><li>Formalised pre-use inspections</li><li>Formal inspection and preventative/ condition-based maintenance system (for TMM)</li><li>Add Periodic UAT for detection accuracy and intervention capability. Define control details</li></ul>                                                         | 2. Unlikely      | 4. High | 3. Mod | -      | 14 (S)        | Y   | N   |
| 3   | CPS L9 - configuration changes not correctly applied                                                             | <ul style="list-style-type: none"><li>CPS L9 SF - Configured collision detection</li><li>CPS L9 SF - Supplier FMECA and product risk assessment</li><li>CPS L9 SF - System training</li><li>Hardware &amp; firmware installation &amp; configuration documentation (for CPS)</li><li>Add Periodic UAT for detection accuracy and intervention capability. Define control details</li></ul>                                                                                                                                                                                                                                                                                                                   | 2. Unlikely      | -       | 2. Min | -      | 5 (L)         | Y   | N   |
| 4   | Operator confidence reduced due to CPS L9 system erratic behavior                                                | <ul style="list-style-type: none"><li>Change management process (for CPS)</li><li>CPS L9 SF - System training</li><li>CPS L9 SF - Configured collision detection</li><li>Installation quality assurance (QA) inspections (for CPS)</li><li>Supplier warranty and performance obligation (for CPS)</li><li>Add CPS L9 SF - Verification of integration with every OEM model TMM where CPS L9 is installed</li><li>Competency-based training (for TMM operators)</li></ul>                                                                                                                                                                                                                                     | 2. Unlikely      | -       | -      | 3. Mod | 9 (M)         | Y   | N   |

## Comprehensive Risk Management Tools:

Provides configurable fields to support the structured identification, evaluation, and mitigation of operational risks.

## Standardised Controls:

Approved control measures are listed and standardized across the system, improving consistency and reducing the potential for misuse or ambiguous control descriptions.

## Internationally Aligned Risk Assessment Tools:

The system supports various formal formats used across mining and industrial operations: Issue-Based Risk Assessments (IBRA), Baseline Risk Assessments (BLRA), Job Risk Analysis (JRA), Bowtie Risk Assessments (BTA), Control Effectiveness Analysis (CEA).

## Standardised Elements for Consistency:

The RMS includes pre-listed controls and standardised unwanted events to enable a more structured and consistent approach to risk documentation. However, users retain flexibility to add new elements, which are flagged for review and approval.

# Example: RMS - IBRA Output Report

TAU5: SURFACE CPS L9 IMPLEMENTATION IBRA & CEA REPORT

DATE: 01/07/2025

VERSION NUMBER: 0.1

## TAU5: SURFACE CPS L9 IMPLEMENTATION IBRA & CEA REPORT

| AUTHORISATIONS | NAME            | POSITION                           | SIGNATURE & DATE |
|----------------|-----------------|------------------------------------|------------------|
| Facilitator    | Quintin Coetzee | IBRA Facilitator / Project Manager |                  |
| Owner          | Annelie Venter  | Automation & Systems Manager       |                  |
| Reviewer       | Thabo Mokoena   | Utilities Engineer                 |                  |
| Approver       | Pieter Lategan  | Senior Engineering Manager         |                  |

\*Items marked with a star were added during the development of this risk assessment.

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Vicinity Detection per TMM regulations.

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## 9. ISSUE BASED RISK ASSESSMENT (IBRA)

### 9.1. PART 1: HAZARD, RISK SOURCE & UNWANTED EVENT IDENTIFICATION

| Nr | Main Area                     | Sub Area                             | Hazard                                                      | Risk Source          | Unwanted Event                                                                                                   | Max Reasonable Consequence Description                                                                                                                                                                                                     | Functional Ownership      |
|----|-------------------------------|--------------------------------------|-------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1  | Engineering Management of CPS | Detection & Intervention Performance | • Large TMM (CPS L9 activated)                              | Mechanical (Mobile)  | TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation | Partial or complete 'loss of control' event. Note: This item is evaluated as a special case to indicate what can go wrong if a CPS L9 functionality is activated at the wrong time.                                                        | Engineering/ CPS Provider |
| 2  | Engineering Management of CPS | Override & Fail-to-Safe Behavior     | • Engineering Personnel<br>• CPS Technicians                | Personal / Behaviour | Personnel bypasses CPS safety functionality                                                                      | Bypass of safety device. Direct consequence is L&R. Indirect follow on consequence is (S) as likelihood of unwanted TMM interaction increases. View taken on (S) rating that UE is closely related to PUE (TMM--TMM unwanted interaction). | Engineering/ CPS Provider |
| 3  | Engineering Management of CPS | Configuration Management             | • CPS L9 - Detection<br>• CPS L9 - Interface & Intervention | Other                | CPS L9 - configuration changes not correctly applied                                                             | Actual setup does not match expected setup. Configuration changes are lost or not applied correctly.                                                                                                                                       | CPS Provider              |
| 4  | Operational Management of CPS | TMM Operator Training & Behavior     | • CPS L9 - Detection<br>• CPS L9 - Interface & Intervention | Personal / Behaviour | Operator confidence reduced due to CPS L9 system erratic behavior                                                | Impact on operator confidence, change in driving style.                                                                                                                                                                                    | CPS Provider              |

### 9.2. PART 2: CURRENT CONTROLS, LIKELIHOOD & MAX RISK RATINGS

| Nr | Unwanted Event                                                                                                   | Current Controls                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Likelihood of UE | (S)     | (H) | (E) | (C) | (L&R) | (M) | (R) | Max Risk Rank |
|----|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|-----|-----|-----|-------|-----|-----|---------------|
| 1  | TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation | <ul style="list-style-type: none"><li>Competency-based training (for TMM operators)</li><li>CPS L9 SF - Detection and intervention system fail to safe (FTS) design</li><li>CPS L9 SF - Ramp Assist (RA) functionality</li><li>Traffic management system - brake test ramps and testing points</li><li>Traffic management system - mine access and mine access routes</li><li>Hardware &amp; firmware installation &amp; configuration documentation (for CPS)</li><li>Formal inspection and preventative/ condition-based maintenance system (for TMM)</li></ul> | 2. Unlikely      | 4: High | -   | -   | -   | -     | -   | -   | 14 (S)        |

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| Nr | Unwanted Event                                                                                                   | Rank   | PUE | AOE | Specific Notes Regarding the Unwanted Event                                                                                                                                                      |
|----|------------------------------------------------------------------------------------------------------------------|--------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation | 14 (S) | Yes | No  | Investigate additional controls if team agrees to rating. Note: This item is evaluated as a special case to indicate what can go wrong if a CPS L9 functionality is activated at the wrong time. |

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